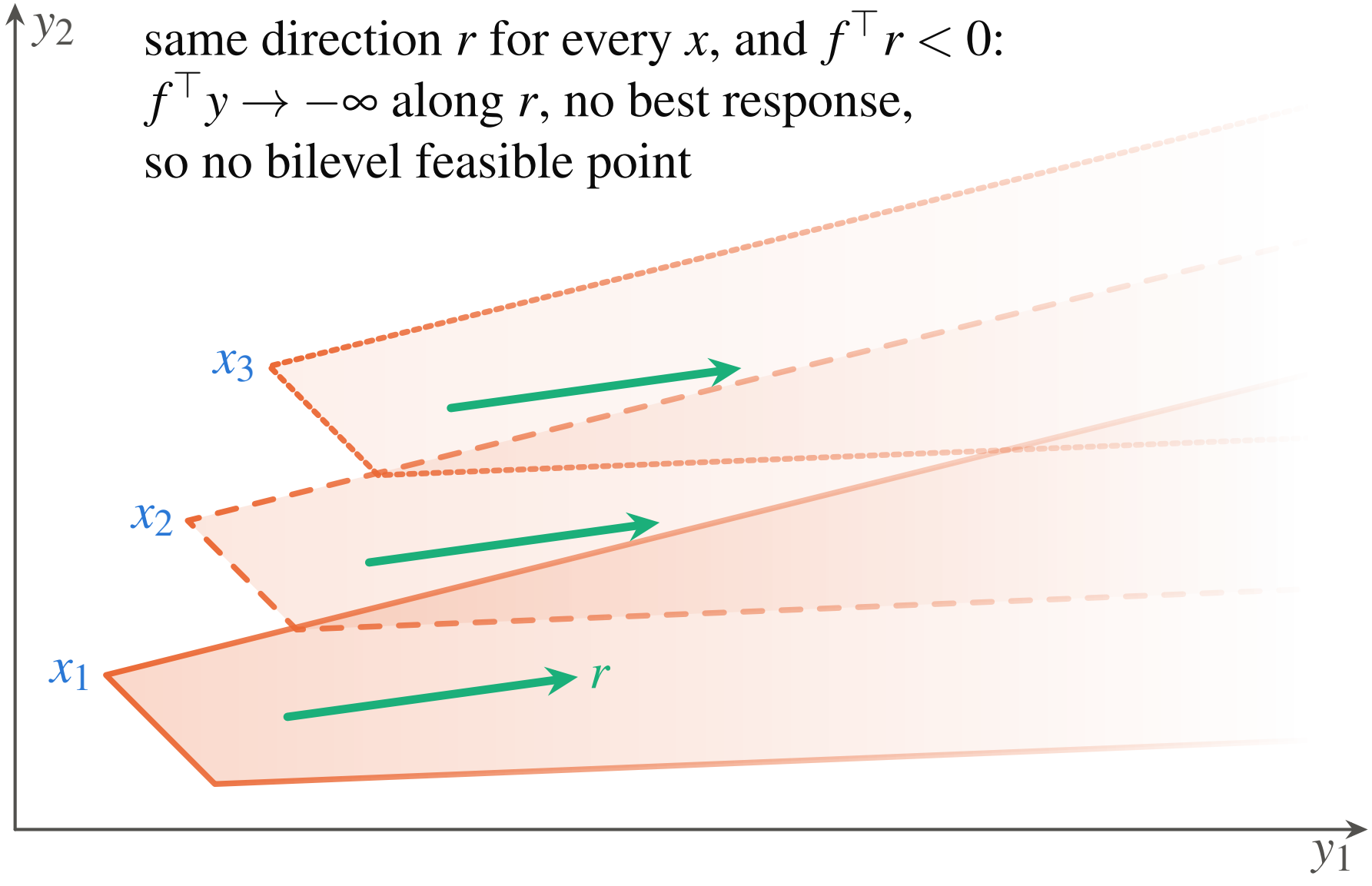
y_2
same direction r for every x, and $f^\top r < 0$:
$f^\top y \to -\infty$ along r, no best response,
so no bilevel feasible point
x_3
x_2
x_1
r
y_1